1. Library Installation and Imports

Installing Libraries

In [144…

```
!pip install emoji
!pip install contractions
```

```
Requirement already satisfied: emoji in /usr/local/lib/python3.11/dist-pac
kages (2.14.1)
Requirement already satisfied: contractions in /usr/local/lib/python3.11/d
ist-packages (0.1.73)
Requirement already satisfied: textsearch>=0.0.21 in /usr/local/lib/python
3.11/dist-packages (from contractions) (0.0.24)
Requirement already satisfied: anyascii in /usr/local/lib/python3.11/dist-
packages (from textsearch>=0.0.21->contractions) (0.3.3)
Requirement already satisfied: pyahocorasick in /usr/local/lib/python3.11/
dist-packages (from textsearch>=0.0.21->contractions) (2.2.0)
```

Importing Libraries

In [145…

```
import pandas as pd
import numpy as np
import re
import emoji
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
import pickle
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.corpus import opinion_lexicon
import unicodedata
import contractions
import matplotlib.pyplot as plt
import seaborn as sns
import string
from wordcloud import WordCloud, STOPWORDS
from collections import Counter
from PIL import Image
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import PCA

nltk.download('punkt')
nltk.download('wordnet')
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))
nltk.download('opinion_lexicon')


# Show all rows and columns
pd.set_option('display.max_rows', None)  # Show all rows
pd.set_option('display.max_columns', None)  # Show all columns
```

```
pd.set_option('display.width', None)  # Adjust width to avoid truncation
pd.set_option('display.max_colwidth', None)  # Show full column content
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package opinion_lexicon to /root/nltk_data...
[nltk_data]   Package opinion_lexicon is already up-to-date!
```

2. Loading Dataset and Initial Exploration

Reading CSV into dataframe

In [146…

```
df = pd.read_csv("./Sentiment_Data.csv", encoding='latin1')
```

Datatypes of dataset

Displaying head and info

In [147…

```
print("Head --" , df.head())
print("-----------------------------------------------------------------------
print("-----------------------------------------------------------------------
print("Info --", df.info())
```

```
Head --
Tweet  \
0             @_angelica_toy Happy Anniversary!!!....The Day the FreeDUM
B Died (In the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freed
umbers #FluTruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1
cIPwmh9
1           @McfarlaneGlenda Happy Anniversary!!!....The Day the FreeDUM
B Died (In the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freed
umbers #FluTruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1
cIPwmh9
2  @thevivafrei @JustinTrudeau Happy Anniversary!!!....The Day the FreeDUM
B Died (In the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freed
umbers #FluTruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1
cIPwmh9
3                 @NChartierET Happy Anniversary!!!....The Day the FreeDUM
B Died (In the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freed
umbers #FluTruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1
cIPwmh9
4           @tabithapeters05 Happy Anniversary!!!....The Day the FreeDUM
B Died (In the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freed
umbers #FluTruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1
cIPwmh9

  Sentiment
0  Mild_Pos
1  Mild_Pos
2  Mild_Pos
3  Mild_Pos
4  Mild_Pos
---------------------------------------------------------------------------
-----------------------------------------------------------
---------------------------------------------------------------------------
-----------------------------------------------------------
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 451332 entries, 0 to 451331
Data columns (total 2 columns):
 #   Column     Non-Null Count   Dtype 
---  ------     --------------   ----- 
 0   Tweet      451331 non-null  object
 1   Sentiment  451332 non-null  object
dtypes: object(2)
memory usage: 6.9+ MB
Info -- None
```

Shape of the dataset

In [148…
```
len(df),df.index.shape[-1]
```

Out[148…
```
(451332, 451332)
```

In [149…
```
print("Shape of dataset: ".format(), df.shape)
```

```
Shape of dataset:  (451332, 2)
```

Extract tweets for each specific sentiment class to inspect sample data

In [150…
```
mild_pos_tweet = df[df['Sentiment'] == 'Mild_Pos']['Tweet']
mild_neg_tweet = df[df['Sentiment'] == 'Mild_Neg']['Tweet']
neutral_tweet = df[df['Sentiment'] == 'Neutral']['Tweet']
strong_pos_tweet = df[df['Sentiment'] == 'Strong_Pos']['Tweet']
strong_neg_tweet = df[df['Sentiment'] == 'Strong_Neg']['Tweet']
```

Group the tweets by sentiment label for easier iteration

In [151…
```
sentiment_samples = {
    "Mild_Pos": mild_pos_tweet,
    "Mild_Neg": mild_neg_tweet,
    "Neutral": neutral_tweet,
    "Strong_Pos": strong_pos_tweet,
    "Strong_Neg": strong_neg_tweet
}
```

3. Exploratory Data Analysis (EDA) - Sentiment Distribution

Print the first 5 tweets of each sentiment category to get a sense of the data

In [152…
```
for label, series in sentiment_samples.items():
    print(f"\n{'='*10} First 5 samples of {label.replace('_', ' ').title(
    for i, tweet in enumerate(series.head(5), 1):
        print(f"{i}. {tweet}\n")
```

```
========== First 5 samples of Mild Pos tweets ==========

1. @_angelica_toy Happy Anniversary!!!....The Day the FreeDUMB Died (In th
e tune of Don McLean's "American Pie") #FreeDumbConvoy #Freedumbers #FluTr
uxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1cIPwmh9

2. @McfarlaneGlenda Happy Anniversary!!!....The Day the FreeDUMB Died (In
the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freedumbers #Flu
TruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1cIPwmh9

3. @thevivafrei @JustinTrudeau Happy Anniversary!!!....The Day the FreeDUM
B Died (In the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freed
umbers #FluTruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1
cIPwmh9

4. @NChartierET Happy Anniversary!!!....The Day the FreeDUMB Died (In the
tune of Don McLean's "American Pie") #FreeDumbConvoy #Freedumbers #FluTrux
Klan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1cIPwmh9

5. @tabithapeters05 Happy Anniversary!!!....The Day the FreeDUMB Died (In
the tune of Don McLean's "American Pie") #FreeDumbConvoy #Freedumbers #Flu
TruxKlan #convoywatch #convoy #FreedomConvoy   https://t.co/ZT1cIPwmh9


========== First 5 samples of Mild Neg tweets ==========

1. @FightHaven Those knee drops remind me of something...

Oh right.
Trudeau's crackdown on the #FreedomConvoy Trucker's protest.

2. @AndreLemelin4 We are a sovereign nation with a democratically elected
government - planting another countryâ€  s flag in front of our federal pa
rliament (particularly in relation to this â€  freedom convoyâ€  ) strongl
y implies treasonous intent!

3. If the freedumb convoy reunion putting an American Flag on Parliament H
ill wasn't enough of a signal this wasn't a Canadian movement to begin wit
h...

Still crying about how we're not a free country and democracy has fallen.

Then again, half think Putin is the leader of freedom.

4. @victoria_misery @CBCNews you support the freedom convoy, but also want
some people forcefully admitted and detained?
And no one but you realized those with mental health and addictions were t
aking up so much housing already. What an insight!

5. @brianlilley Justin Trudeau and his Puppet Masters wanted another Janua
ry 6th. He and the liberal NDP wanted to see violence the peaceful Freedom
trucker Convoy painted as extremist. Absolute disgrace  many of the so-cal
led journalists that are not tellin


========== First 5 samples of Neutral tweets ==========

1. @JustinTrudeau You Belong In Jail.
#VaccineMandates #CrimesAgainstHumanity #TrudeauDictatorship
#FreedomConvoy
```

```
https://t.co/HrsYk2IYXC

2. #FreeDumbConvoy #FreedomConvoy #Freedumbers #freedumb #freedom

3. #FreedomConvoy https://t.co/cF1ohIQ5dK

4. #freedomconvoy https://t.co/levc86wFp3

5. @Cl1Richard Canadians who trust the mainstream media had 4 minutes of n
egative propaganda that if you ask anyone who believes fake news they're r
eaction is more like the freedom convoy was a terrorist group this is what
we're dealing with people are just


========== First 5 samples of Strong Pos tweets ==========

1. Freedom Convoy as InkBlot Test https://t.co/auLrduDpdI

2. @mark_slapinski Well itâ€ s pretty easy to see what their agenda is an
d Pierre has remained silent on the issues and he never actually fought fo
r the convoy just did a photo op

3. @Garnet_2203 @TuckerCarlson @mini_bubbly Your head is so far up Trudeau
â€ s ass you can see what he had for lunch! Trudeau doesnâ€ t debate, he
will never answer an unscripted question! The â€ fringe minorityâ€ was t
he millions who supported the Freedom C

4. @CPC_HQ The Freedom Convoy you supported, which included Diagolon Accel
erationists, cost us billions of dollars. Also your leader is tight with i
t AND #MGTOW. Resign. All of you. You're a disgrace.ð

5. Canadaâ€ s Freedom Radio! hosted by Freedom Convoy 2022 Road Captain:
Jason LaFace https://t.co/78cZOkI5zv on #Podbean


========== First 5 samples of Strong Neg tweets ==========

1. @brethordark The #FreedomConvoy 1 year Anniversary... they don't like F
REEDOM or HONKING!!!

2. #Freedumbers partied as they caused fellow Canadians to be prisoners in
their own homes

When the pandemic came, their inconvenience was much more important, any l
ives lost be damned

But #TrudeauIsAPsychopath? ð ¤

Nah

#FreeDumbConvoy #cdnpoli #FreedomConvoy #TrudeauWasRight

3. have been found guilty repeatedly. But only fines and no criminal charg
es were made.

Create a new society
This is where we are folks. The utopian world of the Liberals. No freedom
of speech C-11, speakers, even the anniversary of the convoy to Ottawa. Th
ese are examples of the
```

```
4. @Humanlty1o1 Oh ffs get over yourselves. REAL heroes stayed on the job
and didn't have the horn honk circle jerk fest that cost us millions. F*ck
the #FreedomConvoy hard. #IStandWithTrudeau

5. "Trudeau must be held accountable and not whitewashed for invoking the
Emergencies Act to punish his political opponents."...READ MORE: https://
t.co/2mVeMCcIOG

OPINION BY @DavidKrayden

#ableg #abpoli #cdnpoli #skpoli #bcpoli #westernstandard #FreedomConvoy #E
mergenciesAct https://t.co/UbKML22rsb
```

Helper function to plot sentiment counts

In [153…

```python
def plot_sentiment_counts(sentiment_counts):
    plt.rcParams['figure.figsize'] = (8, 6)
    bars = plt.bar(sentiment_counts.index, sentiment_counts.values, color
    plt.xlabel('Sentiment Class')
    plt.ylabel('Number of Tweets')
    plt.title('Number of Tweets per Sentiment Class')

    for bar in bars:
        yval = bar.get_height()
        plt.text(bar.get_x() + bar.get_width()/2.0, yval + 500, int(yval)
    plt.show()
```

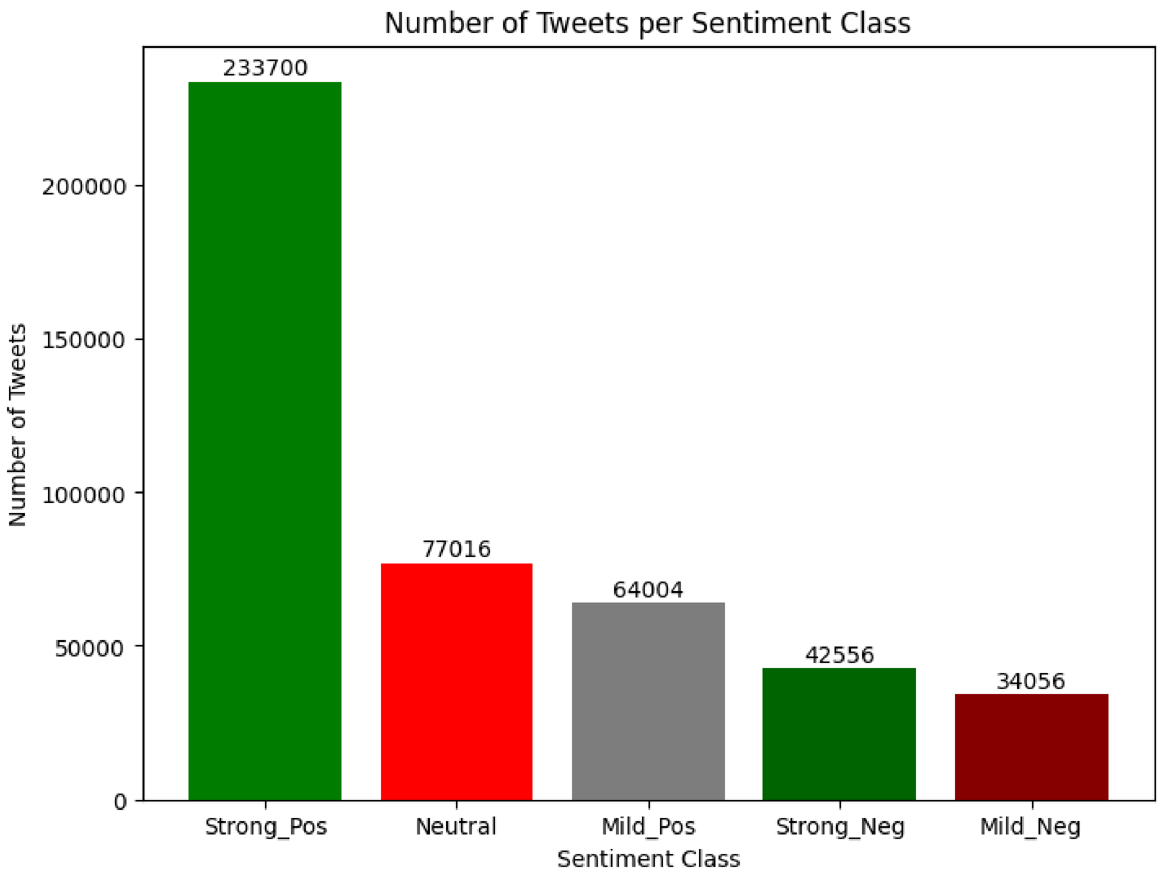
In [154…

```python
# Calculate the number of tweets per sentiment class
sentiment_counts = df['Sentiment'].value_counts()

print('Total Counts of all sentiment classes:\n', sentiment_counts)
print("==============")

# Plot a bar chart to visualize the distribution of tweet counts across s
plot_sentiment_counts(sentiment_counts)
```

```
Total Counts of all sentiment classes:
 Sentiment
Strong_Pos    233700
Neutral        77016
Mild_Pos       64004
Strong_Neg     42556
Mild_Neg       34056
Name: count, dtype: int64
==============
```



4. Feature Extraction - Word Count and Punctuation Count

Calculate the number of words in each tweet and store it in a new column 'word_count'

In [155…

```
# Handles any non-string entries safely by assigning a count of 0
df['word_count'] = df['Tweet'].apply(lambda x: len(str(x).split()) if isi
```

Word count distribution plot

In [156…

```
# Get a sorted list of unique sentiment classes in the dataset
sentiments = sorted(df['Sentiment'].unique())

# Create a 2x3 grid of subplots for plotting word count distributions per
fig, axes = plt.subplots(2, 3, figsize=(18, 8), sharex=True, sharey=True)
axes = axes.flatten()

# For each sentiment class, plot the distribution of word counts using a
for idx, sentiment in enumerate(sentiments):
    word_counts = df[df['Sentiment'] == sentiment]['word_count']
    sns.distplot(word_counts, bins=30, kde=True, color="skyblue", hist_kw
    axes[idx].set_title(f"{sentiment}")
    axes[idx].set_xlabel("Word Count")
    axes[idx].set_ylabel("Density")
    axes[idx].grid(True, linestyle="--", alpha=0.5)
```

```
# Hide any unused subplot if sentiment classes are fewer than subplot slo
if len(sentiments) < len(axes):
    for idx in range(len(sentiments), len(axes)):
        axes[idx].set_visible(False)

# Set the main title and adjust layout for the whole figure
fig.suptitle("Word Count Distribution by Sentiment Class", fontsize=16)
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()
```

```
/tmp/ipython-input-156-2351425197.py:11: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
th
similar flexibility) or `histplot` (an axes-level function for histogram
s).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(word_counts, bins=30, kde=True, color="skyblue", hist_kws=
{"alpha":0.5}, ax=axes[idx])
/tmp/ipython-input-156-2351425197.py:11: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
th
similar flexibility) or `histplot` (an axes-level function for histogram
s).

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  sns.distplot(word_counts, bins=30, kde=True, color="skyblue", hist_kws=
{"alpha":0.5}, ax=axes[idx])
/tmp/ipython-input-156-2351425197.py:11: UserWarning:

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0.

Please adapt your code to use either `displot` (a figure-level function wi
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similar flexibility) or `histplot` (an axes-level function for histogram
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  sns.distplot(word_counts, bins=30, kde=True, color="skyblue", hist_kws=
{"alpha":0.5}, ax=axes[idx])
/tmp/ipython-input-156-2351425197.py:11: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
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Please adapt your code to use either `displot` (a figure-level function wi
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similar flexibility) or `histplot` (an axes-level function for histogram
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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(word_counts, bins=30, kde=True, color="skyblue", hist_kws=
{"alpha":0.5}, ax=axes[idx])
```

```
/tmp/ipython-input-156-2351425197.py:11: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
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similar flexibility) or `histplot` (an axes-level function for histogram
s).

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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(word_counts, bins=30, kde=True, color="skyblue", hist_kws=
{"alpha":0.5}, ax=axes[idx])
```

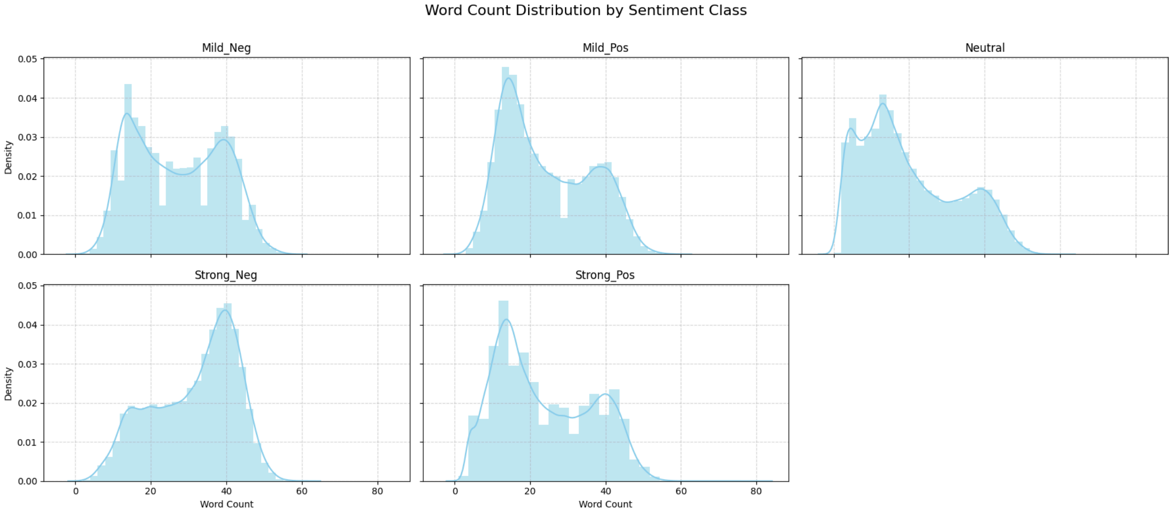

Adding punctuation count for each tweet in the dataframe

In [157…

```
df['punctuation_count'] = df['Tweet'].apply(lambda z: len([c for c in str
```

Punctuation count distribution plot

In [158…

```
sentiments = sorted(df['Sentiment'].unique())
fig, axes = plt.subplots(2, 3, figsize=(18, 8), sharex=True, sharey=True)
axes = axes.flatten()

for idx, sentiment in enumerate(sentiments):
    counts = df[df['Sentiment'] == sentiment]['punctuation_count']
    sns.distplot(counts, bins=20, kde=True, color="purple", hist_kws={"al
    axes[idx].set_title(f"{sentiment}")
    axes[idx].set_xlabel("Punctuation Count")
    axes[idx].set_ylabel("Density")
    axes[idx].grid(True, linestyle="--", alpha=0.5)

for idx in range(len(sentiments), len(axes)):
    axes[idx].set_visible(False)

fig.suptitle("Punctuation Count Distribution by Sentiment Class", fontsiz
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()
```

```
/tmp/ipython-input-158-787223585.py:7: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
th
similar flexibility) or `histplot` (an axes-level function for histogram
s).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(counts, bins=20, kde=True, color="purple", hist_kws={"alph
a":0.5}, ax=axes[idx])
/tmp/ipython-input-158-787223585.py:7: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
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similar flexibility) or `histplot` (an axes-level function for histogram
s).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(counts, bins=20, kde=True, color="purple", hist_kws={"alph
a":0.5}, ax=axes[idx])
/tmp/ipython-input-158-787223585.py:7: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
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similar flexibility) or `histplot` (an axes-level function for histogram
s).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(counts, bins=20, kde=True, color="purple", hist_kws={"alph
a":0.5}, ax=axes[idx])
/tmp/ipython-input-158-787223585.py:7: UserWarning:

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https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(counts, bins=20, kde=True, color="purple", hist_kws={"alph
a":0.5}, ax=axes[idx])
```
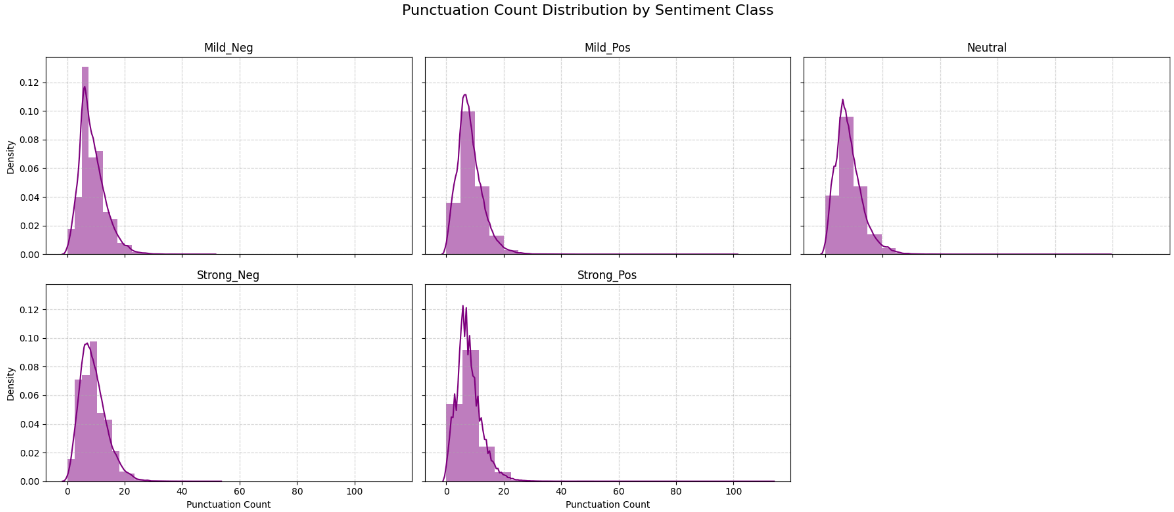

```
/tmp/ipython-input-158-787223585.py:7: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.
0.

Please adapt your code to use either `displot` (a figure-level function wi
th
similar flexibility) or `histplot` (an axes-level function for histogram
s).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(counts, bins=20, kde=True, color="purple", hist_kws={"alph
a":0.5}, ax=axes[idx])
```



5. Handling Missing Values

Checking for nulls

In [159…
```
df.isnull().sum()
```

Out[159…

	0
Tweet	1
Sentiment	0
word_count	0
punctuation_count	0

dtype: int64

Displaying missing rows

In [160…
```
missing_rows = df[df.isnull().any(axis=1)]
print(missing_rows)
```

```
       Tweet Sentiment  word_count  punctuation_count
75986    NaN   Neutral           0                  0
```

Dropping missing values

In [161…
```
df.dropna(inplace=True)
```

Rechecking info and null

In [162…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 451331 entries, 0 to 451331
Data columns (total 4 columns):
 #   Column             Non-Null Count   Dtype 
---  ------             --------------   ----- 
 0   Tweet              451331 non-null  object
 1   Sentiment          451331 non-null  object
 2   word_count         451331 non-null  int64 
 3   punctuation_count  451331 non-null  int64 
dtypes: int64(2), object(2)
memory usage: 17.2+ MB
```

In [163…
```
df.isnull().sum()
```

Out[163…

	0
Tweet	0
Sentiment	0
word_count	0
punctuation_count	0

dtype: int64

6. Text Cleaning Functions and Preprocessing

Helper functions to clean text

In [164…
```
def clean_text(text):
    if not isinstance(text, str):
        return ""
    text = unicodedata.normalize('NFKD', text).encode('ascii', 'ignore').
    text = text.lower()
    text = re.sub(r'<.*?>', ' ', text)
    text = contractions.fix(text)
    text = re.sub(r"https?://\S+|www\.\S+", "", text)
    text = re.sub(r"@\w+", "", text)
    text = re.sub(r"#\w+", "", text)
    text = re.sub(r"[^a-zA-Z\s]", " ", text)
```

```
    text = re.sub(r"\s+", " ", text).strip()
    text = re.sub(r'[^\w\s-]', ' ', text)
    text = re.sub(r'\s+', ' ', text).strip()
    return text
```

Cleaning the tweets by replacing some irregular texts

In [165…
```
df['clean_tweet'] = df['Tweet'].apply(clean_text)
```

In [166…
```
def demojize_text(text):
    text = emoji.demojize(text, delimiters=(" ", " "))
    emoji_pattern = re.compile("["
        u"\U0001F600-\U0001F64F"
        u"\U0001F300-\U0001F5FF"
        u"\U0001F680-\U0001F6FF"
        u"\U0001F1E0-\U0001F1FF"
        u"\U00002702-\U000027B0"
        u"\U000024C2-\U0001F251"
        "]+", flags=re.UNICODE)

    emoji_count = len(emoji_pattern.findall(text))

    text = emoji_pattern.sub(r'', text)
    return text, emoji_count
```

Cleaning the tweets by replacing emoticons

In [167…
```
df[['clean_tweet', 'emoji_count']] = df['clean_tweet'].apply(
    lambda x: pd.Series(demojize_text(x))
)

# Now get the total emoji handled
total_emoji = df['emoji_count'].sum()
print(f"Emoji Handled: {total_emoji}")
```

```
Emoji Handled: 0
```

Slang dictionary and slang replacement function

In [168…
```
slang_dict = {
    # Common chat abbreviations / slang
    "idk": "i do not know",
    "brb": "be right back",
    "smh": "shaking my head",
    "tbh": "to be honest",
    "omg": "oh my god",
    "lol": "laughing out loud",
    "lmao": "laughing my ass off",
    "fyi": "for your information",
    "btw": "by the way",
    "imo": "in my opinion",
    "imho": "in my humble opinion",
    "ikr": "i know right",
    "ttyl": "talk to you later",
    "bff": "best friends forever",
    "afk": "away from keyboard",
```

```
    "rofl": "rolling on the floor laughing",
    "np": "no problem",
    "ftw": "for the win",
    "jk": "just kidding",
    "ppl": "people",
    "ur": "your",
    "cya": "see you",
    "thx": "thanks",
    "gr8": "great",
    "pls": "please",
    "plz": "please",

    # Numbers/text shorthand
    "u": "you",
    "n": "and",
    "r": "are",
    "4": "for",
    "2": "to",
    "b4": "before",
    "l8r": "later",
    "tho": "though",

    # Additional common informal/slang
    "yw": "you are welcome",
    "bc": "because",
    "b/c": "because",
    "cuz": "because",
    "w/": "with",
    "w/o": "without",
    "afaik": "as far as i know",
    "fomo": "fear of missing out",
    "irl": "in real life",
    "wtf": "what the fuck",
    "idc": "i do not care",
    "tmi": "too much information",
    "omw": "on my way",
    "gg": "good game",
    "ggwp": "good game well played",
    "dm": "direct message"
}
```

Function to replace slang words in a text with their full forms using the slang_dict

In [169…

```
def replace_slang(text, slang_dict= slang_dict):
    words = text.split()
    words = [slang_dict.get(w, w) for w in words]
    slang_count = sum(1 for w in words if w in slang_dict)
    return " ".join(words), slang_count
```

Cleaning the tweets by replacing slang words

In [170…

```
df[['clean_tweet', 'slang_count']] = df['clean_tweet'].apply(
    lambda x: pd.Series(replace_slang(x))
)
```

```
total_slang = df['slang_count'].sum()
print(f"Slang Handled: {total_slang}")
```

```
Slang Handled: 0
```

7. Negation Handling

Defining negation words

In [171…

```
negation_words = set(['not', 'no', 'never', "n't", 'cannot', 'neither', '
```

In [172…

```
def negate_text(text):
    tokens = text.split()
    negated_tokens = []
    negate = False
    negation_scope = 3  # number of words after negation to tag

    i = 0
    while i < len(tokens):
        word = tokens[i]
        if word in negation_words:
            negate = True
            negated_tokens.append(word)
            i += 1
            # Tag next few words
            for j in range(negation_scope):
                if i < len(tokens):
                    negated_tokens.append(tokens[i] + "_NEG")
                    i += 1
            negate = False
        else:
            negated_tokens.append(word)
            i += 1
    return ' '.join(negated_tokens)
```

In [173…

```
# Apply negation handling to clean_tweet column
df['clean_tweet'] = df['clean_tweet'].apply(negate_text)

# Optionally, count how many negations were handled (e.g., count how many
def count_negations(text):
    return text.count('_NEG')

df['negation_count'] = df['clean_tweet'].apply(count_negations)

total_negations = df['negation_count'].sum()
print(f"Negations Handled: {total_negations}")
```

```
Negations Handled: 330955
```

8. Tokenization and Lemmatization

Tokenizing the words

In [174...
```
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

Out[174... True

In [175...
```
from nltk.tokenize import word_tokenize
df['tokens'] = df['clean_tweet'].apply(word_tokenize)
```

In [176...
```
df['tokens'] = df['tokens'].apply(lambda x: [w for w in x if w not in sto
```

Lemmatizing the tokens

In [177...
```
lemmatizer = WordNetLemmatizer()
df['tokens'] = df['tokens'].apply(lambda x: [lemmatizer.lemmatize(w) for
```

In [178...
```
df['clean_tweet'] = df['tokens'].apply((lambda x: ' '.join(x)))
```

In [179...
```
print(df.columns)
```

```
Index(['Tweet', 'Sentiment', 'word_count', 'punctuation_count', 'clean_twe
et',
       'emoji_count', 'slang_count', 'negation_count', 'tokens'],
      dtype='object')
```

9. Word Cloud and Frequent Words Visualization

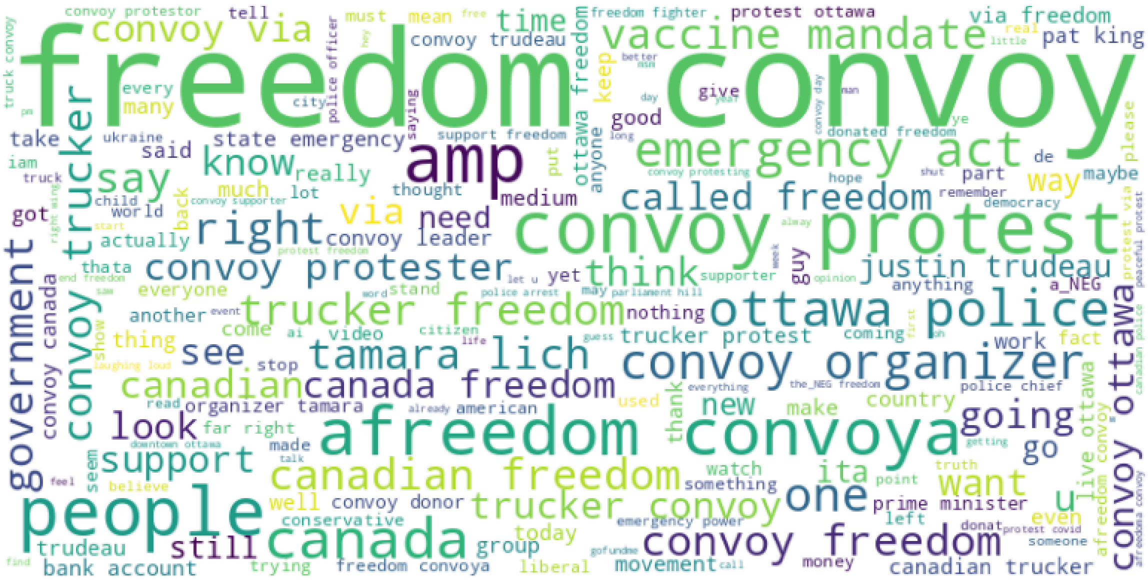
Word cloud for 5 classes (Mild_Neg, Strong_Neg, Neutral, Mild_Pos, Strong_Pos)

In [180...
```
classes = sorted(df['Sentiment'].unique())

img_path = None

for sentiment in classes:
    texts = df[df['Sentiment'] == sentiment]['clean_tweet']
    if img_path is not None:
        mask = np.array(Image.open(img_path))
    else:
        mask = None
    wc = WordCloud(
        stopwords=STOPWORDS,
        mask=mask,
        background_color="white",
        contour_width=2,
        contour_color="black",
        max_words=200,
        max_font_size=256,
```

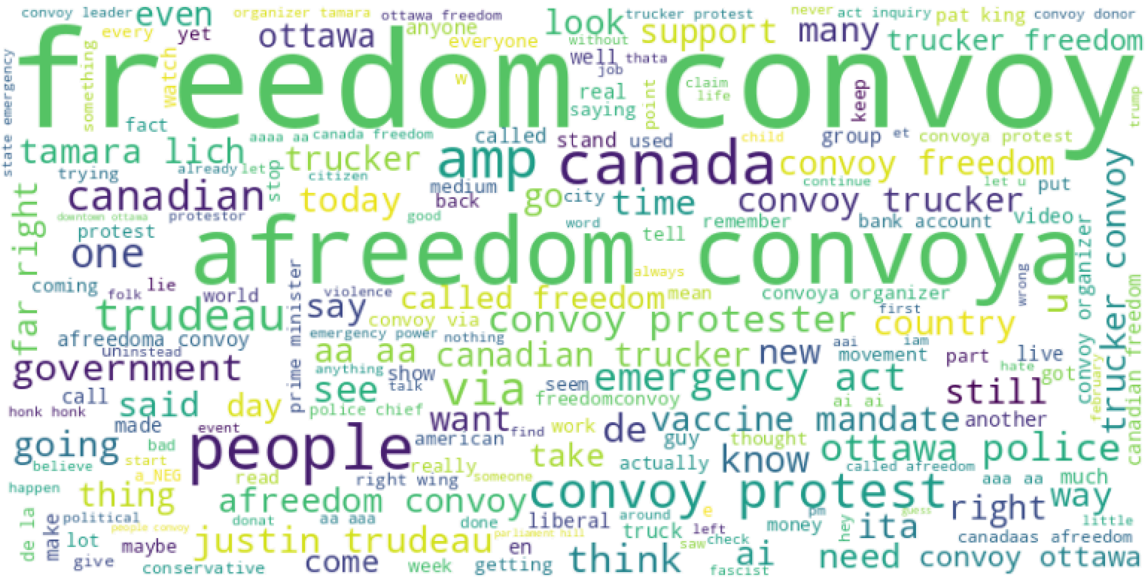
```
        random_state=42,
        width=800 if mask is None else mask.shape[1],
        height=400 if mask is None else mask.shape[0]
    ).generate(' '.join(map(str, texts.dropna())))
    plt.figure(figsize=(8, 5))
    plt.imshow(wc, interpolation="bilinear")
    plt.axis('off')
    plt.title(f"Word Cloud: {sentiment}", fontsize=16)
    plt.tight_layout()
    plt.show()
```

Word Cloud: Mild_Neg

Word Cloud: Mild_Pos



Word Cloud: Neutral



Word Cloud: Strong_Neg

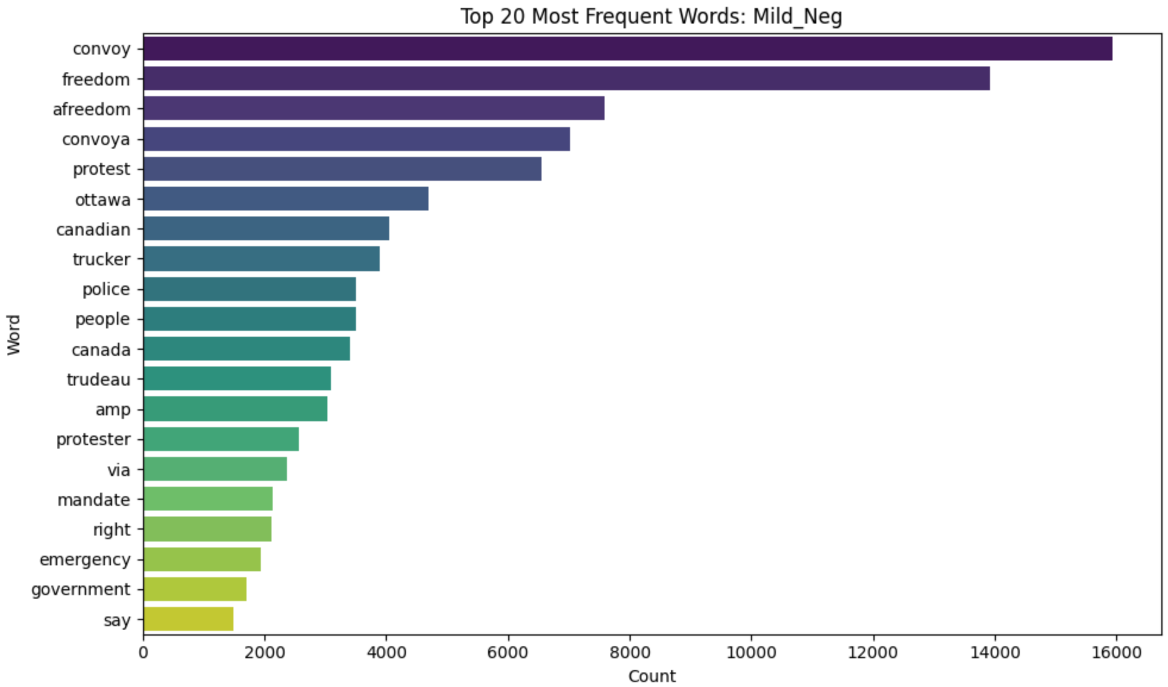
Word Cloud: Strong_Pos

Top 20 most frequent words for all 5 classes

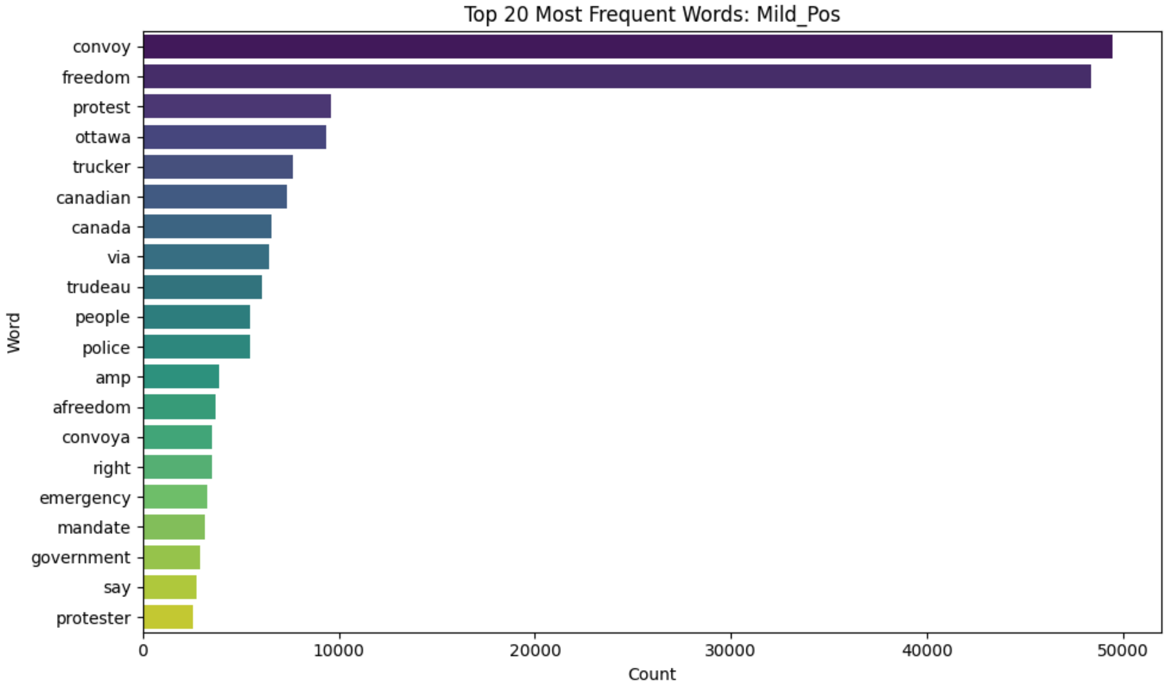
In [181…

```
def plot_top_words(texts, n=20, title=None):
    words = [
        word.lower() for text in texts
        for word in word_tokenize(text)
        if word.lower() not in STOPWORDS and len(word) > 2 and word.isalp
    ]
    word_counts = Counter(words)
    top_words = word_counts.most_common(n)
    plt.figure(figsize=(10, 6))
    sns.barplot(
        x=[count for word, count in top_words],
        y=[word for word, count in top_words],
        palette='viridis'
    )
    plt.title(title or f'Top {n} Most Frequent Words')
    plt.xlabel('Count')
    plt.ylabel('Word')
    plt.tight_layout()
    plt.show()

# For all classes
for sentiment in classes:
    texts = df[df['Sentiment'] == sentiment]['clean_tweet']
    plot_top_words(
        texts, n=20, title=f"Top 20 Most Frequent Words: {sentiment}"
    )
```

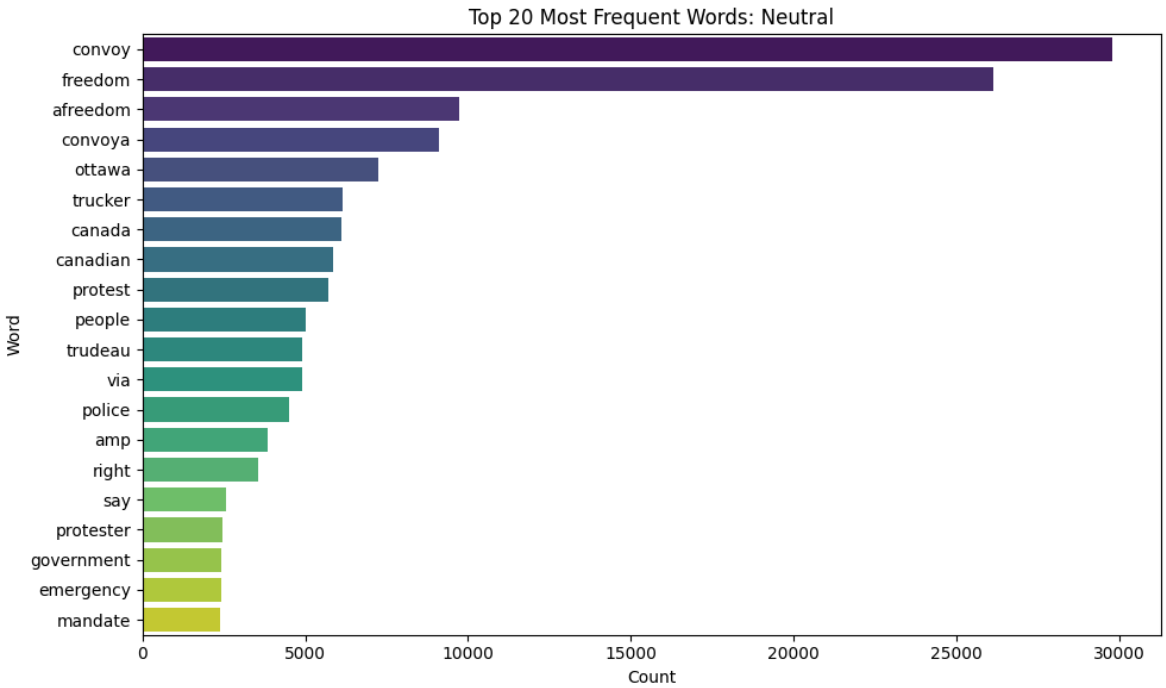
```
/tmp/ipython-input-181-4107960595.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for
the same effect.

  sns.barplot(
```



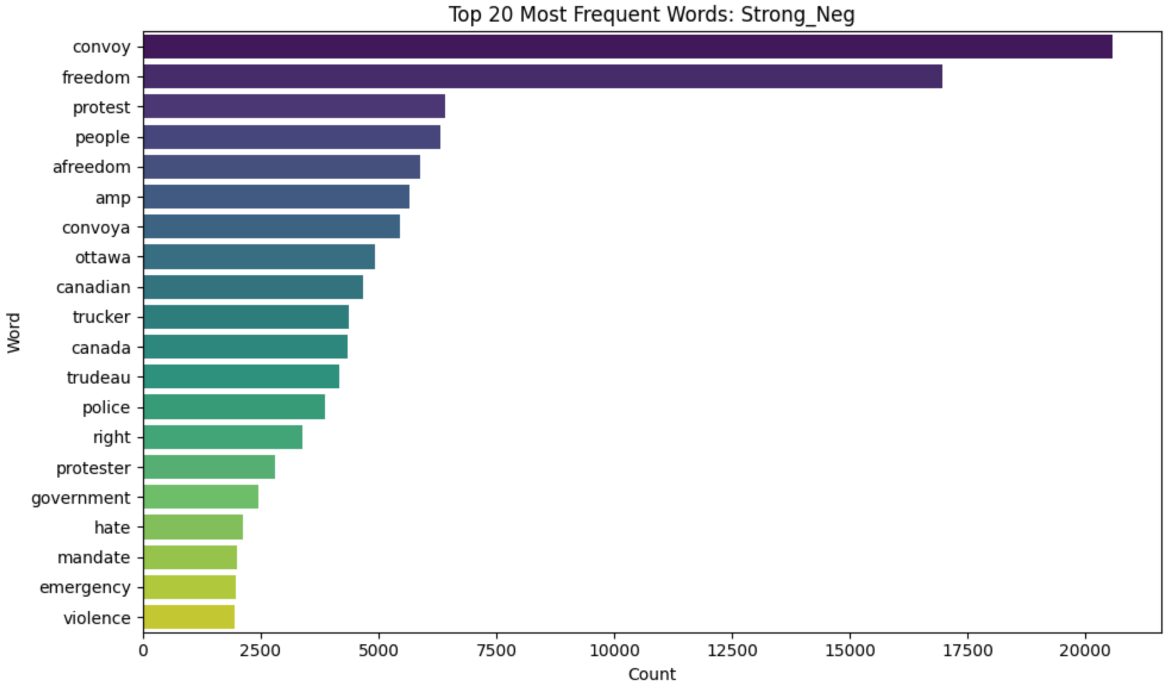
```
/tmp/ipython-input-181-4107960595.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for
the same effect.

  sns.barplot(
```



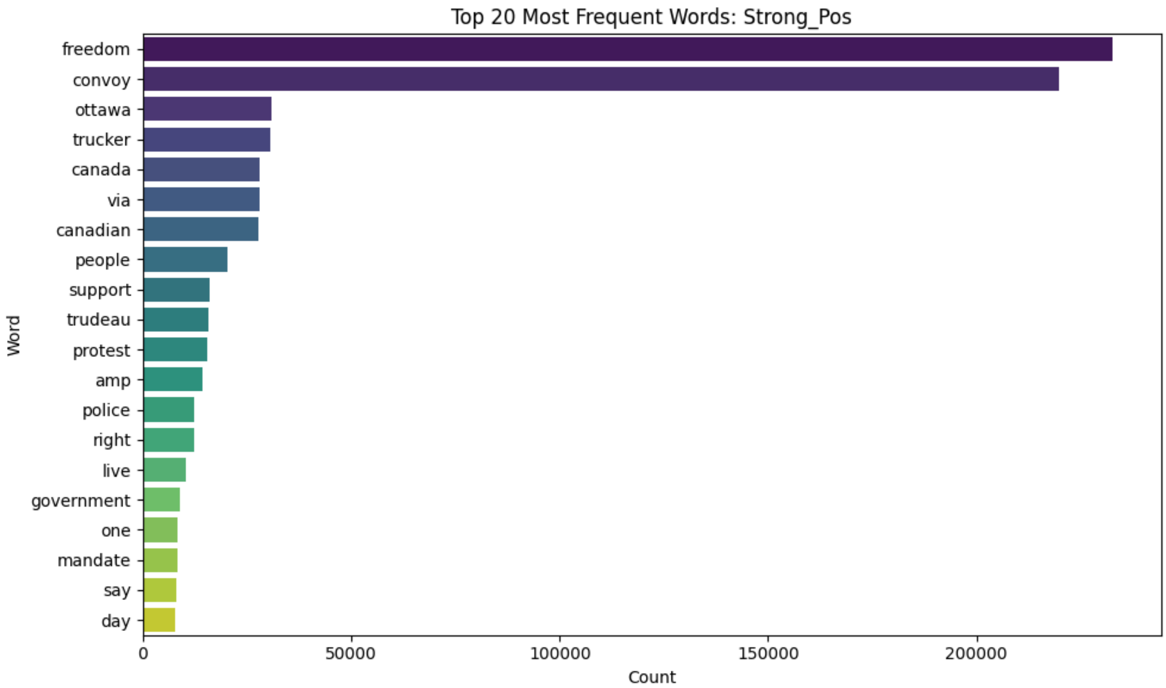

```
/tmp/ipython-input-181-4107960595.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for
the same effect.

  sns.barplot(
```



```
/tmp/ipython-input-181-4107960595.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for
the same effect.

  sns.barplot(
```




```
/tmp/ipython-input-181-4107960595.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be remove
d in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for
the same effect.

  sns.barplot(
```



10. Removing Empty Tweets After Cleaning

In [182…

```
total_tweet = len(df)
empt_str = df['clean_tweet'].eq('').sum()
print(f"Total tweets: {total_tweet:,}")
empt_str_removed = empt_str
print(f"Total empty strings (after cleaning): {empt_str_removed}")
```

```
Total tweets: 451,331
Total empty strings (after cleaning): 10270
```

Removing the row from dataframe which has empty string after cleaning the data

In [183…

```
df = df[df['clean_tweet'].str.strip().astype(bool)]
```

In [184…

```
total_tweet = len(df)
empt_str = df['clean_tweet'].eq('').sum()
print(f"Total tweets: {total_tweet:,}")
print(f"Total empty strings (after cleaning): {empt_str:,}")
```

```
Total tweets: 441,061
Total empty strings (after cleaning): 0
```

11. Consolidating Sentiment Classes

Mapping original 5 sentiment labels into 3 classes: negative, neutral, positive

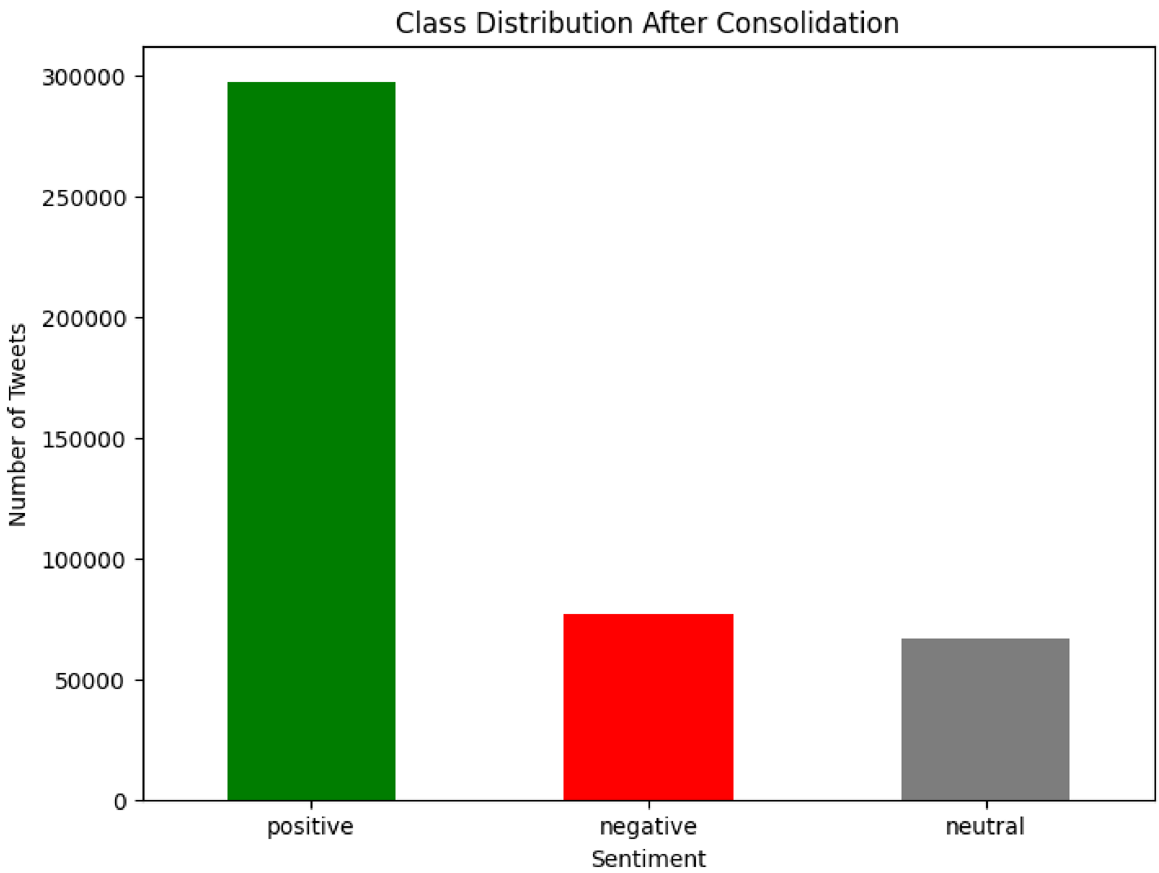
In [185…

```
def consolidate_label(x):
    if x in ['Mild_Neg', 'Strong_Neg']:
        return 'negative'
    elif x in ['Mild_Pos', 'Strong_Pos']:
        return 'positive'
    else:
        return 'neutral'

df['sentiment_consolidated'] = df['Sentiment'].apply(consolidate_label)
```

Plotting distribution after consolidation

In [186…

```
df['sentiment_consolidated'].value_counts().plot(kind='bar', color=['gree
plt.title('Class Distribution After Consolidation')
plt.xlabel('Sentiment')
plt.ylabel('Number of Tweets')
plt.xticks(rotation=0)
plt.show()
```



Printing counts and proportions

In [187…

```
print("Counts:")
print(df['sentiment_consolidated'].value_counts())

print("\nProportions:")
print(df['sentiment_consolidated'].value_counts(normalize=True))
```

```
Counts:
sentiment_consolidated
positive    297457
negative     76569
neutral      67035
Name: count, dtype: int64

Proportions:
sentiment_consolidated
positive    0.674412
negative    0.173602
neutral     0.151986
Name: proportion, dtype: float64
```

There is class imbalance in the dataset as the tweet with sentiment positive are more than the negative and neutral.

12. Train / Validation / Test Split

(before Feature engineering, tf-idf, PCA, Lexicon Score Feature and Label Encoding to avoid Data Leakage)

In [188…

```
from sklearn.model_selection import train_test_split

# Step 1: Define inputs
X_full = df[['Tweet', 'clean_tweet']]
y_full = df['sentiment_consolidated']

# Step 2: First split – Separate test set (20%)
X_temp, X_test, y_temp, y_test = train_test_split(
    X_full, y_full, test_size=0.2, stratify=y_full, random_state=42
)

# Step 3: Second split – From remaining, get train (60%) and val (20%)
X_train, X_val, y_train, y_val = train_test_split(
    X_temp, y_temp, test_size=0.25, stratify=y_temp, random_state=42
)  # 0.25 x 0.8 = 0.2

# Step 4: Reset indices
X_train = X_train.reset_index(drop=True)
X_val = X_val.reset_index(drop=True)
X_test = X_test.reset_index(drop=True)

y_train = y_train.reset_index(drop=True)
y_val = y_val.reset_index(drop=True)
y_test = y_test.reset_index(drop=True)

# ✅ Final Proportions: 60% Train / 20% Val / 20% Test
```

13. Feature Engineering: Tweet Length and Hashtag Count

In [189…

```
# Feature 1: Tweet Length (based on original tweet text)
X_train['tweet_length'] = X_train['Tweet'].apply(len)
X_val['tweet_length'] = X_val['Tweet'].apply(len)
X_test['tweet_length'] = X_test['Tweet'].apply(len)

# Feature 2: Hashtag Count (using regex)
import re
X_train['num_hashtags'] = X_train['Tweet'].apply(lambda x: len(re.findall
X_val['num_hashtags'] = X_val['Tweet'].apply(lambda x: len(re.findall(r"#
X_test['num_hashtags'] = X_test['Tweet'].apply(lambda x: len(re.findall(r
```

14. TF-IDF Vectorization and PCA Visualization

In [190…

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Vectorization (fit only on training data)
tf_vectorizer = TfidfVectorizer(max_features=500)
X_train_vec = tf_vectorizer.fit_transform(X_train['clean_tweet'])
X_val_vec   = tf_vectorizer.transform(X_val['clean_tweet'])
```

```
X_test_vec  = tf_vectorizer.transform(X_test['clean_tweet'])

# 2. PCA (fit only on training data)
pca = PCA(n_components=2)
X_train_pca = pca.fit_transform(X_train_vec.toarray())
X_val_pca   = pca.transform(X_val_vec.toarray())
X_test_pca  = pca.transform(X_test_vec.toarray())

# 3. Visualize PCA (train set only)
df_pca = pd.DataFrame({
    'PC1': X_train_pca[:, 0],
    'PC2': X_train_pca[:, 1],
    'sentiment': y_train
})

plt.figure(figsize=(8, 6))
sns.scatterplot(data=df_pca, x='PC1', y='PC2', hue='sentiment', palette='
plt.title('PCA: Class Separability of Training Data')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend(title='Sentiment')
plt.show()
```

15. Lexicon Score Feature

In [191…
```
pos_words = set(opinion_lexicon.positive())
neg_words = set(opinion_lexicon.negative())
```

In [192…
```
def lexicon_score(text, pos_words, neg_words):
    tokens = text.split()
```

```
    pos = sum(1 for w in tokens if w in pos_words)
    neg = sum(1 for w in tokens if w in neg_words)
    return pos - neg
```

In [193…

```
# Apply lexicon_score feature to all three sets safely
X_train['lexicon_score'] = X_train['clean_tweet'].apply(lambda x: lexicon
X_val['lexicon_score'] = X_val['clean_tweet'].apply(lambda x: lexicon_sco
X_test['lexicon_score'] = X_test['clean_tweet'].apply(lambda x: lexicon_s
```

16. Label Encoding

In [194…

```
# ================================
# Label Encoding (Post-Split, No Leakage)
# ================================
from sklearn.preprocessing import LabelEncoder

# Initialize and fit only on training labels
label_encoder = LabelEncoder()
y_train_encoded = label_encoder.fit_transform(y_train)
y_val_encoded   = label_encoder.transform(y_val)
y_test_encoded  = label_encoder.transform(y_test)

# Optional: to check mapping
label_mapping = dict(zip(label_encoder.classes_, label_encoder.transform(
print("Label Encoding Mapping:", label_mapping)
```

```
Label Encoding Mapping: {'negative': np.int64(0), 'neutral': np.int64(1),
'positive': np.int64(2)}
```

In [195…

```
label_encoder.classes_
```

Out[195…

```
array(['negative', 'neutral', 'positive'], dtype=object)
```

In [196…

```
df.tail()
```

Out[196…

	Tweet	Sentiment	word_count	punctuation_count	clea
451327	Gaza; Peace n' Freedom - Viva Palestina convoy enters Algeria: \r\nThe Viva Palestina convoy, the massive relief c.. http://tinyurl.com/cajylt	Strong_Pos	19	12	gaz freec p alg p
451328	Face of Defense: Soldier Finds Freedom in U.S., Fights for Freedom in Iraq: CONVOY SUPPORT CEN... http://tinyurl.com/dyth62 RT @freerepublic	Strong_Pos	19	14	face sol freec sup
451329	Face of Defense: Soldier Finds Freedom in U.S., Fights for Freedom in Iraq: CONVOY SUPPORT CENTER SCANIA, Iraq, .. http://tinyurl.com/dyth62	Strong_Pos	20	14	face sol freec sc
451330	Gaza; Peace n' Freedom - "Israel stops aid convoy en route to Gaza from Hebron": \r\nMa"an News Agency report.. http://tinyurl.com/7mguhc	Strong_Pos	20	15	gaz isı aic gaza qu
451331	@convoy 83 yes! get on freedom server!	Strong_Pos	7	3	

17. Exporting Processed Data and Model Artifacts

Data Preprocessing Checklist

In [201…

```
# 1. Regex usage check (approximate by counting regex calls in clean_text
print("- Regex Cleaning: Applied in clean_text function")

# 2. Emoji Handling
total_emoji = df['emoji_count'].sum() if 'emoji_count' in df.columns else
print(f"- Emoji Handling: Done, total emojis handled = {total_emoji}")

# 3. Slang Handling
total_slang = df['slang_count'].sum() if 'slang_count' in df.columns else
print(f"- Slang Handling: Done, total slang replacements = {total_slang}"

# 4. Abbreviation Handling (if combined with slang_dict, same as slang_co
total_abbr = df['abbr_count'].sum() if 'abbr_count' in df.columns else 0
print(f"- Abbreviation Handling: {'Done' if total_abbr>0 else 'Not separa

# 5. Negation Handling
total_negations = df['negation_count'].sum() if 'negation_count' in df.co
print(f"- Negation Handling: {'Done' if total_negations>0 else 'Not detec

# 6. Empty tweets removed
empty_count = df['clean_tweet'].eq('').sum()
print(f"- Empty Tweets Removed: {'Yes, ' + str(empt_str_removed) + ' remo

# 7. Class distribution after consolidation
if 'sentiment_consolidated' in df.columns:
    print("- Sentiment Classes Distribution:")
    print(df['sentiment_consolidated'].value_counts())
else:
    print("- Sentiment Classes Distribution: Not found")

# 8. Train/Val/Test Split Sizes (if variables exist)
try:
    print(f"- Train size: {len(X_train)}")
    print(f"- Validation size: {len(X_val)}")
    print(f"- Test size: {len(X_test)}")
except NameError:
    print("- Train/Val/Test Split: Variables not found")

# 9. Feature Columns Present (tweet_length, num_hashtags, lexicon_score)
features = ['tweet_length', 'num_hashtags', 'lexicon_score']
for feat in features:
    present = feat in df.columns or ('X_train' in globals() and feat in X
    print(f"- Feature '{feat}': {'Present' if present else 'Missing'}")

print("========================================")
```

```
- Regex Cleaning: Applied in clean_text function
- Emoji Handling: Done, total emojis handled = 0
- Slang Handling: Done, total slang replacements = 0
- Abbreviation Handling: Not separated from slang, total abbreviations rep
laced = 0
- Negation Handling: Done, total negations tagged = 330955
- Empty Tweets Removed: Yes, 10270 removed, remaining empty tweets = 0
- Sentiment Classes Distribution:
sentiment_consolidated
positive    297457
negative     76569
neutral      67035
Name: count, dtype: int64
- Train size: 264636
- Validation size: 88212
- Test size: 88213
- Feature 'tweet_length': Present
- Feature 'num_hashtags': Present
- Feature 'lexicon_score': Present
========================================
```

Exporting train, val, test sets (features and labels) as CSV

In [198…

```
import os

os.makedirs('../data/processed', exist_ok=True)

X_train.to_csv('X_train.csv', index=False)
X_val.to_csv('X_val.csv', index=False)
X_test.to_csv('X_test.csv', index=False)

y_train.to_frame().to_csv('y_train.csv', index=False)
y_val.to_frame().to_csv('y_val.csv', index=False)
y_test.to_frame().to_csv('y_test.csv', index=False)
```

Exporting the label encoder

In [202…

```
import pickle
with open('label_encoder1.pkl', 'wb') as f:
    pickle.dump(label_encoder, f)
```

In [ ]: